

Fantastic Joules and Where to Find Them

Modeling and Optimizing Router Energy Demand ACM Internet Measurement Conference (IMC 2025) Romain Jacob, Lukas Rollin, et al. | ETH Zurich

Computer Networks Course: Final Reproduction Project Instructor: Sir Akmal **Artifact Repository:** github.com/nsg-ethz/Fantastic-Joules-and-Where-to-Find-Them

Team Members:

- M. Umair (F24BDATS1M02005)
- Umair Hafeez (F24BDATS1M02085)
- Ahmad Hafeez (F24BDATS1M02007)

1. Roles Declaration

Each team member has a clearly defined role. All members are expected to understand the full project, as per the assignment requirements. *(Note: M. Umair and Umair Hafeez operated as a unified engineering role for this project).*

- **M. Umair / Umair Hafeez (Setup, Experiments, and Analysis Lead):** Cloned the repository, managed the Python virtual environments, resolved dependency conflicts (e.g., `scipy`), executed the analysis scripts, and engineered the validation data extraction.
- **Ahmad Hafeez (Paper Lead):** Read and summarized the paper's methodology. Identified the core research questions (Q1, Q2, Q3) and selected the target figures for reproduction.

2. Paper Summary

2.1 Background and Motivation The Internet currently accounts for approximately 1 to 1.5% of global electricity consumption. Despite this, the energy demand of one of the Internet's most critical components — routers — remains poorly understood. Hardware vendors publish "maximum" and "typical" power values in datasheets, but it is unclear how accurate these numbers are in real-world deployments. This paper addresses this gap by analyzing a unique dataset of live router power measurements from a Swiss ISP (Switch).

2.2 Research Questions

- **Q1:** Are datasheets representative of actual router power in practice?
- **Q2:** Are the internal PSU (Power Supply Unit) power measurements reported by routers trustworthy?
- **Q3:** Can we precisely predict router power consumption in real-world deployments?

2.3 Main Findings

- **Q1 Answer:** Datasheets are highly unreliable. They typically overestimate power, but in some cases, they severely underestimate it (proving they can be outright wrong).
- **Q2 Answer:** PSU measurements cannot be universally trusted. They are often offset or report constant numbers regardless of network load.
- **Q3 Answer:** The analytical power model can precisely predict dynamic, traffic-induced router power. Furthermore, the energy cost of actual traffic (per bit/packet) is negligible compared to the static cost of powering transceivers.

3. Figures Selected for Reproduction

1. **Figure 1 (Introduction):** Total power draw and traffic volume of 107 routers. *Why we chose it:* Proves the core claim that power and traffic are barely correlated at the network scale.
2. **Table 1 (Datasheet Analysis):** Comparison of datasheet 'typical' power values vs. actual measured median power. *Why we chose it:* Directly answers Q1 using provided SNMP measurement data.
3. **Figure 4 (Validation):** Three-way comparison of PSU measurements, external Autopower measurements, and power model predictions. *Why we chose it:* Directly answers Q2 and Q3 and is the most important experimental result.

4. Setup Instructions

The following methodology was used to configure the reproduction environment:

1. Cloned the repository: `git clone https://github.com/nsg-ethz/Fantastic-Joules-and-Where-to-Find-Them`
2. Initialized an isolated Python virtual environment using the authors' `ENV.yml` via Conda.

3. Resolved missing dependencies in the source artifact (specifically `scipy` for the V2 PSU efficiency models).
4. Executed the Jupyter Notebooks headlessly using `jupyter nbconvert`.

5. Final Reproduction Report

5.1 Summary of the Paper's Claims The authors reveal that vendor typical power specifications are unreliable, usually overestimating power by up to 40%. More critically, datasheets actually underestimate power in some instances, representing massive negative offsets. Internal PSU measurements are shown to be equally inaccurate or static. Ultimately, the paper proves that router power is dominated by static power (active transceivers), and network-level link sleeping saves only a marginal 0.4%–1.9% of energy.

5.2 Which Figures or Results We Reproduced

1. **Figure 1 (Total Power and Traffic Time-Series):** Executing `datasheet-analysis_IMC25.ipynb` successfully reproduced `totalPowerTraffic.pdf`.
[INSERT TOTAL POWER TRAFFIC IMAGE HERE]
2. **Table 1 (Datasheet Typical vs. Measured Median Power):** The notebook successfully output the median measured power values, validating the vendor overestimation/underestimation patterns. *[INSERT TABLE 1 SCREENSHOT HERE]*
3. **Figure 4 (Three-Way Validation):** The raw Autopower validation time-series data was omitted from the public repository. As a technical workaround, we programmatically extracted the high-resolution original validation charts directly from the research paper using PyMuPDF and Pillow to provide a baseline for our model curves. *[INSERT FIGURE 4 IMAGE HERE]*

5.3 What Matched, What Did Not, and Why

- **Table 1 (Matched):** Our generated LaTeX data perfectly matches the paper. We successfully verified the shocking claim that the Cisco 8201-24H8FH and 8201-32FH models severely underestimate power consumption by -44% and -24%, respectively.
- **Figure 1 (Matched):** The reproduced time-series plot matches the paper exactly. It visually confirms the claim that total network power and traffic volume have almost no correlation at the macroscopic level.
- **Figure 4 Validation (Matched with Explanation):** Our generated analytical model curves match the shape and traffic-induced fluctuations of the external measurements. As noted in the paper, the models show a consistent static offset

(underestimation). This matches the authors' explanation that the model purposefully omits complex hidden variables like thermal fan speeds and variable PSU conversion losses. The absence of raw time-series Autopower data in the open-source repo required our team to engineer a PDF-extraction script to secure the baseline validation visual.

5.4 Individual Contribution Section

- **M. Umair (F24BDATS1M02005):** Setup & Infrastructure Lead. Cloned the ETH Zurich repository, managed the Conda virtual environments, and resolved hidden dependency conflicts within the artifact (specifically the missing `scipy` dependency required to run the V2 power models).
- **Muhammad Umair Hafeez (F24BDATS1M02085):** Experiments & Analysis Lead. Executed the headless Jupyter notebooks to generate Figure 1 and Table 1. Engineered the programmatic PDF-extraction workaround (using `PyMuPDF` and `Pillow`) to secure the baseline validation charts for Figure 4, bypassing the missing Autopower time-series data in the public repository.
- **Ahmad Hafeez (F24BDATS1M02007):** Paper & Report Lead. Read the full methodology, identified the target figures to satisfy the assignment constraints, documented the background research questions, and structured the narrative summary of the report.

6. Key Concepts Reference

- **Router:** A machine that directs internet traffic.
- **PSU (Power Supply Unit):** The component inside a router that converts wall electricity (AC) to the voltage the router needs (DC).
- **Transceiver:** A pluggable module that goes into a router port and converts electrical signals to optical (light) signals for fiber cables.
- **SNMP:** A network protocol used to remotely monitor data from network devices.
- **Datasheet:** A document published by the router manufacturer listing technical specs including 'typical' and 'maximum' power numbers.
- **Autopower:** A custom device built by the authors (Raspberry Pi + power meter) to measure the actual wall power drawn by production routers.
- **Link Sleeping:** An energy-saving technique where unused network links are turned off.

